AUTOMATICITY OF HANKEL DETERMINANTS OF THE BINARY SUM OF DIGITS MODULO POWERS OF TWO

KNOWLEDGE SENSE

Abstract. Let $s(j)$ be the number of ones in the binary expansion of j, and let $D(N) = \det[s(i+j)]_{0\le i,j<N}$. We prove that for every integer $K \ge 1$ the sequence $(D(n+1) \bmod 2^K)_{n\ge 0}$ is 2-automatic. This answers a question of Sobolewski and Ulas. The proof starts from their recursion for Hankel determinants of weighted binary sums of digits. After a suitable normalization, the auxiliary bordered determinant in this recursion becomes an odd integer and the ratio of the two determinants has odd denominator, so the recursion can be reduced modulo 2^K. The reflected indices produced by the recursion correspond to complemented suffixes of the binary expansion, and the reduced recursion is carried out by a deterministic finite automaton that reads the expansion from the least significant digit. We also observe that $(D(N))_{N\ge 1}$ itself is not 2-regular.

1. Introduction

Let $s(j)$ denote the number of ones in the binary expansion of an integer $j \ge 0$, so that $s(0) = 0$, and put

$$D(N) = \det\big[s(i+j)\big]_{0\le i,j<N} \qquad (N \ge 1).$$

The sequence $(D(N))_{N\ge 1}$ is A056886 in the OEIS [10] and begins

$$0,\ -1,\ 2,\ 5,\ -8,\ 3,\ 8,\ 141,\ -204,\ 11,\ -4,\ -11,\ -30,\ 99,\ 1836,\ 164997,\ \ldots$$

Allouche and Shallit asked for a simple closed form for these determinants [3, p. 208, Problem 4]. Hankel determinants of automatic sequences have been studied since the work of Allouche, Peyrière, Wen and Wen on the Thue–Morse sequence [1], where the automaticity of related determinants reduced modulo 2 was established. Later results of this kind include the work of Guo and Wen on difference sequences of the Thue–Morse sequence [7], Han's Hankel determinant calculus [8], and the theorem of Hu and Han that the reduced Hankel determinants modulo 2 of a family of ± 1-valued automatic sequences are automatic [9]. Exact evaluations are rarer; see Fokkink, Kraaikamp and Shallit [5] for the period-doubling sequence. Guo proved that the Hankel determinant sequence of the characteristic sequence of the powers of two is 2-regular [6, Theorem 3], and he determined explicit values of related determinants at special indices [6, Theorem 4].

Sobolewski and Ulas [11] embedded s into the family of weighted binary sums of digits and obtained a recursion [11, Theorem 4.2] that computes $D(N)$ for every

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N. Among many other results, they proved that $D(N) \equiv N + 1 \pmod 2$ [11, Corollary 4.14], and they asked the following question.

Question ([11, Question 4.15])**.** Let $k \geq 1$. Is the sequence $(D(n+1) \bmod 2^k)_{n\geq 0}$ 2-automatic?

They report that computations suggest an affirmative answer for $k = 2$ and $k = 3$, with automata of 12 and 83 states, respectively. Our main result answers this question in the affirmative for every k.

Theorem 1.1. *For every integer $K \geq 1$, the sequence $(D(n+1) \bmod 2^K)_{n\geq 0}$ is 2-automatic.*

For $K = 1$ the sequence is periodic by [11, Corollary 4.14], so the new content concerns $K \geq 2$. The automaton we construct depends on K and is explicit; the upper bound on its number of states that the construction provides is very large (Remark 6.5). The question of a simple closed form for $D(N)$ is not addressed here.

Theorem 1.1 cannot be deduced from 2-regularity of $(D(N))_{N\geq 1}$ itself. The residues of a 2-regular integer sequence modulo a fixed integer form a 2-automatic sequence [2, Corollary 2.4], but $(D(N))_{N\geq 1}$ grows exponentially along $N = 2^m$ and is therefore not 2-regular (Proposition 7.1).

Outline of the proof. For $2^L < N \leq 2^{L+1}$ the recursion of [11] expresses the determinant $D(N)$, together with an auxiliary bordered determinant, in terms of the corresponding pair at a smaller order N'; either $N' = 2^{L+1} - N + 1$ or $N' = N - 2^L$. The price is a change of weights: at each stage all weights are 1 except possibly the top weight, which has the form 2^{-r}. Two points must be addressed. First, the recursion divides by $2^{-r} - 2$ and contains powers of 2, so it is not clear that it survives reduction modulo 2^K. We show that after multiplying the bordered determinant by $2^{re(N)}$, where $e(N) = 2N - 2^{L+1} - 1$, it becomes an odd integer, while the ratio of the two determinants lies in the ring $\mathbb{Z}_{(2)}$ of rationals with odd denominator (Proposition 4.1). Second, the reflection $N \mapsto 2^{L+1} - N + 1$ is not a shift of binary digits. However, on $n = N-1$ it complements all binary digits, so the whole recursion amounts to reading the binary expansion of n from the left while keeping one orientation bit (Lemma 5.1). The data needed at each step depend on the remaining suffix only through its length, capped at $\max(K+1, 3)$, and its value modulo 2^K. Reading from the right, these data and the effect of the remaining part of the recursion can be maintained in a finite set of states (Section 6).

Dependencies. In the proof of Theorem 1.1, the only result that we use without proof is the recursion of Sobolewski and Ulas [11, Theorem 4.2], in the specialization [11, Proposition 4.3] with $t = 1$ (Proposition 3.1 below), and only for orders $N \geq 5$. The determinants of orders at most four are evaluated directly (Lemma 3.2), and the case $K = 1$ is covered by the same construction. We do not use the product formula [11, Proposition 3.7] for the bordered determinants; the two facts we need about them, nonvanishing and the 2-adic size of the normalization, are derived from the recursion itself. From the theory of automatic sequences we use only the definition, together with standard equivalences [3] when passing between conventions (Remark 6.4). Proposition 7.1, which is not used in the proof of Theorem 1.1, relies in addition on the evaluation of $D(2^m)$ in [11, Proposition 4.5] and on the growth bound for regular sequences [2, Theorem 2.10].

Organization. Section 2 fixes notation. Section 3 recalls the recursion of Sobolewski and Ulas in the form needed here and treats small orders. The 2-adic

normalization is proved in Section 4. In Section 5 the recursion is unfolded along the binary expansion and reduced modulo 2^K. Section 6 constructs the automaton and proves Theorem 1.1. Section 7 contains further remarks and describes accompanying finite computations.

2. Preliminaries

2.1. **Binary words.** A word will mean a finite sequence of symbols from $\{0,1\}$, and ε denotes the empty word. For a word $v = v_1 \cdots v_\ell$ we write $|v| = \ell$ and $[v]_2 = \sum_{i=1}^{\ell} v_i 2^{\ell-i}$, with $[\varepsilon]_2 = 0$. The complement $\bar{v}$ is obtained by exchanging 0 and 1 in every position, so that

$$[\bar{v}]_2 = 2^{|v|} - 1 - [v]_2. \tag{2.1}$$

For $o \in \{0,1\}$ put $v^{(0)} = v$ and $v^{(1)} = \bar{v}$. For $a, b \in \{0,1\}$, $a \oplus b$ denotes addition modulo 2. For $n \geq 1$, $\mathrm{bin}(n)$ is the binary expansion of n without leading zeros, so its first letter is 1; we put $\mathrm{bin}(0) = \varepsilon$.

2.2. **Residues modulo powers of two.** Let $\mathbb{Z}_{(2)} \subset \mathbb{Q}$ be the ring of rationals with odd denominator. For $K \geq 1$ put $P = 2^K$ and $A = \mathbb{Z}/P\mathbb{Z}$. Reduction modulo P extends to a ring homomorphism $\mathbb{Z}_{(2)} \to A$, and an element of $\mathbb{Z}_{(2)}$ is a unit if and only if its numerator is odd; correspondingly, the units of A are the residues of odd integers.

Lemma 2.1. *If u is an odd integer, then $u^{2^{K-1}} \equiv 1 \pmod{2^K}$. In particular $u^P \equiv 1 \pmod{P}$, so for $E \in A$ the residue $u^E \in A$, computed with any nonnegative representative of E, is well defined.*

Proof. For $K = 1$ the claim is $u \equiv 1 \pmod 2$. If $u^{2^{K-1}} = 1 + 2^K a$ with $a \in \mathbb{Z}$, then $u^{2^K} = 1 + 2^{K+1}a + 2^{2K}a^2 \equiv 1 \pmod{2^{K+1}}$. □

2.3. **Automatic sequences.** A 2-DFAO is a tuple $M = (S, \delta, s_0, \Delta, \tau)$ consisting of a finite set S of states, a transition map $\delta\colon S \times \{0,1\} \to S$, an initial state $s_0 \in S$, an output alphabet Δ and an output map $\tau\colon S \to \Delta$. We let M read words from right to left: put $\hat{\delta}(s, \varepsilon) = s$ and $\hat{\delta}(s, vb) = \hat{\delta}(\delta(s, b), v)$ for a word v and a letter b. A sequence $(a(n))_{n\geq 0}$ with values in Δ is 2-*automatic* if there is a 2-DFAO M such that

$$a(n) = \tau\big(\hat{\delta}(s_0, w)\big) \qquad \text{for all } n \geq 0 \text{ and all words } w \text{ with } [w]_2 = n.$$

Thus the automaton reads the least significant digit first, as in [9], and is insensitive to leading zeros. This notion coincides with the one in which the most significant digit is read first, and with finiteness of the 2-kernel $\{(a(2^i n + j))_{n\geq 0} : i \geq 0,\ 0 \leq j < 2^i\}$; see [3, Section 5.2 and Theorem 6.6.2] and [4, Proposition V.3.3]. For the automaton constructed below the finiteness of the 2-kernel is also immediate (Remark 6.4).

3. The recursion of Sobolewski and Ulas

For a sequence of weights $\mathbf{w} = (w_j)_{j\geq 0}$ and an integer $n = \sum_j n_j 2^j$ with $n_j \in \{0,1\}$, the weighted sum of digits is $s_{\mathbf{w}}(n) = \sum_j n_j w_j$, and $d_{\mathbf{w}}(m) = s_{\mathbf{w}}(m+1) - s_{\mathbf{w}}(m)$ [11].

For $N \geq 2$ let $L = L(N)$ be the integer with $2^L < N \leq 2^{L+1}$, and put

$$e(N) = 2N - 2^{L+1} - 1.$$

Then $e(N)$ is a positive odd integer. Given an indeterminate x, consider the weights $w_0 = \cdots = w_L = 1$ and $w_{L+1} = x$. Only $s_{\mathbf{w}}(m)$ with $m \le 2N-2 < 2^{L+2}$ occur below, so the remaining weights play no role. Define

$$h(N,x) = \det\big[s_{\mathbf{w}}(i+j)\big]_{0\le i,j<N}, \qquad g(N,x) = \det \widetilde{G}(N,x),$$

where $\widetilde{G}(N,x)$ is the $N\times N$ matrix whose entry in position (i,j) is $d_{\mathbf{w}}(i+j)$ for $0 \le j \le N-2$ and 1 for $j = N-1$. Both are polynomials in x with integer coefficients. In the notation of [11, Section 4.1] we have $h(N,x) = h(N,1,x)$ and $g(N,x) = \tilde{g}(N,1,x)$; that is, g is the bordered determinant $\widetilde{G}_{\mathbf{w}}(N)$ of [11, Section 3], obtained from the Hankel matrix of $d_{\mathbf{w}}$ by replacing its last column with a column of ones. For $x = 1$ all relevant weights equal 1, so

$$D(N) = h(N,1) \qquad (N \ge 2). \tag{3.1}$$

Throughout, $\eta_N = (-1)^N$. The following is the case $t = 1$ of [11, Proposition 4.3], which in turn is the specialization of [11, Theorem 4.2] to the weights $(1, t, \dots, t^k, x)$.

Proposition 3.1 (Sobolewski–Ulas [11, Proposition 4.3])**.** *Let $N \ge 5$ and $L = L(N)$. The following identities hold in $\mathbb{Q}(x)$.*

(a) *If $2^L < N \le 3\cdot 2^{L-1}$, put $N' = 2^{L+1} - N + 1$. Then*

$$h(N,x) = \eta_N (x-2)^{e(N)}\Big(-h(N',1) + \frac{\eta_N}{x-2}\, g(N',1)\Big),$$
$$g(N,x) = \eta_N (x-2)^{e(N)}\, g(N',1).$$

(b) *If $3\cdot 2^{L-1} < N \le 2^{L+1}$, put $N' = N - 2^L$ and $b = 2N - 3\cdot 2^L - 1$. Then*

$$h(N,x) = 2^b (x-2)^{2^L}\Big(h\big(N', \tfrac{x}{2}\big) + \frac{\eta_N}{x-2}\, g\big(N', \tfrac{x}{2}\big)\Big),$$
$$g(N,x) = 2^b (x-2)^{2^L}\, g\big(N', \tfrac{x}{2}\big).$$

In both cases $2^{L-1} < N' \le 2^L$, so $L(N') = L-1$; in case (b) *moreover $e(N') = b$.*

In case (a) the top weight of the smaller determinant is $w_L = 1$, which is why h and g are evaluated at 1 on the right-hand side; in case (b) it is $x/2$. The last assertions are elementary: $N \le 3\cdot 2^{L-1}$ gives $N' \ge 2^{L-1}+1$, $N > 2^L$ gives $N' \le 2^L$ in case (a), and $e(N') = 2(N-2^L) - 2^L - 1 = b$ in case (b). In the proof of [11, Proposition 4.1], on which [11, Theorem 4.2] rests, the case distinction for the entries of the block H_l should read $i + j < 2^k$ instead of $i + j < 2^m$; this does not affect any statement.

Lemma 3.2. *We have*

$$h(2,x) = -1, \qquad g(2,x) = 2 - x,$$
$$h(3,x) = 3 - x, \qquad g(3,x) = 2 - x,$$
$$h(4,x) = (x-2)(3x-8), \qquad g(4,x) = (x-2)^2(x-4).$$

Proof. For $N = 2$ we have $L = 0$, the values $s_{\mathbf{w}}(0), s_{\mathbf{w}}(1), s_{\mathbf{w}}(2)$ are $0, 1, x$, and $d_{\mathbf{w}}(0) = 1$, $d_{\mathbf{w}}(1) = x - 1$; hence

$$h(2,x) = \det\begin{bmatrix} 0 & 1 \\ 1 & x \end{bmatrix} = -1, \qquad g(2,x) = \det\begin{bmatrix} 1 & 1 \\ x-1 & 1 \end{bmatrix} = 2 - x.$$

For $N = 3, 4$ we have $L = 1$, and the values $s_{\mathbf{w}}(0), \dots, s_{\mathbf{w}}(6)$ are $0, 1, 1, 2, x, x+1, x+1$, so $d_{\mathbf{w}}(0), \dots, d_{\mathbf{w}}(5)$ are $1, 0, 1, x-2, 1, 0$. For $N = 3$,

$$h(3,x) = \det \begin{bmatrix} 0 & 1 & 1 \\ 1 & 1 & 2 \\ 1 & 2 & x \end{bmatrix} = 3 - x, \qquad g(3,x) = \det \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & x-2 & 1 \end{bmatrix} = 2 - x.$$

For $N = 4$ the two matrices are

$$\begin{bmatrix} 0 & 1 & 1 & 2 \\ 1 & 1 & 2 & x \\ 1 & 2 & x & x+1 \\ 2 & x & x+1 & x+1 \end{bmatrix}, \qquad \begin{bmatrix} 1 & 0 & 1 & 1 \\ 0 & 1 & x-2 & 1 \\ 1 & x-2 & 1 & 1 \\ x-2 & 1 & 0 & 1 \end{bmatrix}.$$

In the first one, interchange the first two rows and clear the first column below the new pivot; expanding along the first column gives

$$h(4,x) = -\det \begin{bmatrix} 1 & 1 & 2 \\ 1 & x-2 & 1 \\ x-2 & x-3 & 1-x \end{bmatrix} = -\det \begin{bmatrix} x-3 & -1 \\ -1 & 5-3x \end{bmatrix} = 3x^2 - 14x + 16,$$

which equals $(x-2)(3x-8)$. In the second one, subtract the first row from the third and $x-2$ times the first row from the fourth; expanding along the first column and then along the row $(x-2, 0, 0)$ gives

$$\begin{aligned} g(4,x) &= \det \begin{bmatrix} 1 & x-2 & 1 \\ x-2 & 0 & 0 \\ 1 & 2-x & 3-x \end{bmatrix} = -(x-2)\big((x-2)(3-x) - (2-x)\big) \\ &= (x-2)^2(x-4). \end{aligned}$$

□

The polynomials $h(4,1,y) = (y-2)(3y-8)$ and $\tilde{g}(4,1,y) = (y-2)^2(y-4)$ are also recorded in the proof of [11, Proposition 4.5].

Corollary 3.3. *For all $N \geq 2$ and $r \geq 0$ we have $g(N, 2^{-r}) \neq 0$.*

Proof. By induction on N, simultaneously for all $r \geq 0$. For $N \leq 4$ this follows from Lemma 3.2, since $0 < 2^{-r} \leq 1$. For $N \geq 5$, Proposition 3.1 expresses $g(N, 2^{-r})$ as a nonzero multiple of $g(N', 1)$ or of $g(N', 2^{-r-1})$, with $2 \leq N' < N$, because $2^{-r} - 2 \neq 0$. □

4. A 2-adic normalization

For $N \geq 2$ and $r \geq 0$ put $q = 2^r$ and define

$$Q(N,r) = q^{e(N)}\, g(N, q^{-1}), \qquad R(N,r) = \frac{h(N,q^{-1})}{g(N,q^{-1})}.$$

By Corollary 3.3, $R(N,r)$ is a well-defined rational number.

Proposition 4.1. *Let $r \geq 0$, $q = 2^r$ and $u = 1 - 2q$.*

(i) *For $N \in \{2, 3, 4\}$,*

$$\begin{aligned} &Q(2,r) = 2q - 1, && R(2,r) = \frac{q}{1-2q}, \\ &Q(3,r) = 2q - 1, && R(3,r) = \frac{1-3q}{1-2q}, \\ &Q(4,r) = (1-2q)^2(1-4q), && R(4,r) = \frac{q(3-8q)}{(1-2q)(1-4q)}. \end{aligned} \tag{4.1}$$

(ii) *Let $N \geq 5$, $L = L(N)$ and $2^L < N \leq 3 \cdot 2^{L-1}$, and let $N' = 2^{L+1} - N + 1$. Then*

$$Q(N,r) = \eta_N \, u^{e(N)} \, Q(N',0), \qquad R(N,r) = -R(N',0) + \eta_N \frac{q}{u}. \tag{4.2}$$

(iii) *Let $N \geq 5$, $L = L(N)$ and $3 \cdot 2^{L-1} < N \leq 2^{L+1}$, and let $N' = N - 2^L$. Then*

$$Q(N,r) = u^{2^L} \, Q(N', r+1), \qquad R(N,r) = R(N', r+1) + \eta_N \frac{q}{u}. \tag{4.3}$$

(iv) *For all $N \geq 2$ and $r \geq 0$, $Q(N,r)$ is an odd integer and $R(N,r) \in \mathbb{Z}_{(2)}$.*

(v) *For all $N \geq 2$, $D(N) = Q(N,0)\, R(N,0)$.*

Proof. Put $x = q^{-1}$, so that $q(x-2) = u$ and $(x-2)^{-1} = q/u$.

(i) We have $e(2) = e(3) = 1$ and $e(4) = 3$. The formulas follow from Lemma 3.2 by multiplying $g(N,x)$ by $q^{e(N)}$ and dividing $h(N,x)$ by $g(N,x)$; for instance

$$q^3(x-2)^2(x-4) = (1-2q)^2(1-4q), \qquad \frac{3x-8}{(x-2)(x-4)} = \frac{q(3-8q)}{(1-2q)(1-4q)}.$$

(ii) Multiplying the second identity of Proposition 3.1(a) by $q^{e(N)}$ gives $Q(N,r) = \eta_N u^{e(N)} g(N',1)$, and $g(N',1) = Q(N',0)$. Dividing the first identity by the second gives $R(N,r) = -R(N',0) + \eta_N (x-2)^{-1}$.

(iii) Here $e(N) = 2^L + e(N')$ and $x/2 = (2q)^{-1}$. Multiplying the second identity of Proposition 3.1(b) by $q^{e(N)} = q^{2^L} q^{e(N')}$ and using $2^{e(N')} q^{e(N')} = (2q)^{e(N')}$ gives

$$Q(N,r) = \big(q(x-2)\big)^{2^L} (2q)^{e(N')} g\big(N', (2q)^{-1}\big) = u^{2^L} Q(N', r+1).$$

Dividing the first identity by the second gives the formula for $R(N,r)$.

(iv) We argue by induction on N, simultaneously for all r. For $N \leq 4$ the integers $2q-1$ and $(1-2q)^2(1-4q)$ are odd and the denominators $1-2q$ and $(1-2q)(1-4q)$ in (4.1) are odd. For $N \geq 5$, the smaller order N' in (4.2) and (4.3) satisfies $2 \leq N' < N$, the factors $\pm u^{e(N)}$ and u^{2^L} are odd integers, and $\eta_N q/u \in \mathbb{Z}_{(2)}$.

(v) For $r = 0$ we have $q = 1$, hence $Q(N,0) = g(N,1)$, and $D(N) = h(N,1) = g(N,1)R(N,0)$ by (3.1). □

Remark 4.2. The normalizing factor $q^{e(N)}$ is the one suggested by [11, Proposition 3.7], according to which $g(N,x)$ is, up to sign, a product of $N-1$ factors of the form $w_l - 2^{l-j} w_j$ with $0 \leq j < l \leq L+1$, exactly $e(N)$ of which have $l = L+1$. Proposition 4.1 obtains the consequences needed here directly from Proposition 3.1 and Lemma 3.2.

5. Unfolding the recursion along the binary expansion

Lemma 5.1. *Let $N \geq 5$ and $w = w_1 \cdots w_\ell = \mathrm{bin}(N-1)$. Then $\ell \geq 3$ and $L(N) = \ell - 1$, and N satisfies $N \leq 3 \cdot 2^{L-1}$ if and only if $w_2 = 0$. Let N' be as in Proposition 3.1. If $w_2 = 0$, then $\mathrm{bin}(N'-1) = \bar{w}_2 \cdots \bar{w}_\ell$. If $w_2 = 1$, then $\mathrm{bin}(N'-1) = w_2 \cdots w_\ell$.*

Proof. The inequalities $2^L < N \leq 2^{L+1}$ mean $2^L \leq N-1 < 2^{L+1}$, that is, $\ell = L+1$; and $N \geq 5$ means $N-1 \geq 4$, that is, $\ell \geq 3$. Next, $N \leq 3 \cdot 2^{L-1}$ holds if and only if $N - 1 < 3 \cdot 2^{\ell-2}$, that is, if and only if $w_2 = 0$. If $w_2 = 0$, then by (2.1), $N' - 1 = 2^\ell - 1 - (N-1) = [\bar{w}]_2$, and $\bar{w} = 0\, \bar{w}_2 \cdots \bar{w}_\ell$ with $\bar{w}_2 = 1$. If $w_2 = 1$, then $N' - 1 = (N-1) - 2^{\ell-1} = [w_2 \cdots w_\ell]_2$. □

Thus the reflection in Proposition 3.1(a) complements the remaining digits, and both branches delete one digit. This leads to the following bookkeeping.

Definition 5.2. Let $n \geq 1$ and $\mathrm{bin}(n) = b_1 \cdots b_t$, and put $k = \max(1, t-1)$. Define $o_i \in \{0,1\}$ and $r_i \geq 0$ for $1 \leq i \leq k$ by $o_1 = 0$, $r_1 = 0$ and, for $1 \leq i < k$,

$$\chi_i = b_{i+1} \oplus o_i, \qquad (o_{i+1}, r_{i+1}) = \begin{cases} (1-o_i,\, 0) & \text{if } \chi_i = 0, \\ (o_i,\, r_i + 1) & \text{if } \chi_i = 1. \end{cases}$$

Put $N_i = 1 + [(b_i \cdots b_t)^{(o_i)}]_2$ and $\ell_i = t - i + 1$.

Lemma 5.3. *In Definition 5.2, for $1 \leq i \leq k$ we have $b_i \oplus o_i = 1$, so $(b_i \cdots b_t)^{(o_i)} = \mathrm{bin}(N_i - 1)$ has length ℓ_i. Moreover $N_1 = n+1$ and $N_k \in \{2,3,4\}$. For $1 \leq i < k$ we have $N_i \geq 5$; the order N_i is in case* (a) *of Proposition 3.1 if and only if $\chi_i = 0$, and N_{i+1} is the order N_i' attached to N_i there.*

Proof. We have $i < k$ if and only if $\ell_i \geq 3$, while $\ell_k \leq 2$. The statements follow by induction on i from Lemma 5.1, applied to $w = (b_i \cdots b_t)^{(o_i)}$, whose second letter is χ_i. In case (a), $\mathrm{bin}(N_i' - 1)$ is the complement of $(b_{i+1} \cdots b_t)^{(o_i)}$, that is, $(b_{i+1} \cdots b_t)^{(1-o_i)}$; in case (b) it is $(b_{i+1} \cdots b_t)^{(o_i)}$. Finally, $\ell_k \leq 2$ and $b_k \oplus o_k = 1$ give $N_k \in \{2,3,4\}$. □

Proposition 5.4. *In the setting of Definition 5.2, put $q_i = 2^{r_i}$, $u_i = 1 - 2q_i$ and, for $1 \leq i < k$,*

$$c_i = \eta_{N_i} \frac{q_i}{u_i}, \qquad (a_i, f_i) = \begin{cases} \big(-1,\ \eta_{N_i} u_i^{e(N_i)}\big) & \text{if } \chi_i = 0, \\ \big(1,\ u_i^{2^{\ell_i - 1}}\big) & \text{if } \chi_i = 1. \end{cases}$$

Define $\alpha_1 = 1$, $\beta_1 = 0$, $\gamma_1 = 1$ and $\alpha_{i+1} = \alpha_i a_i$, $\beta_{i+1} = \beta_i + \alpha_i c_i$, $\gamma_{i+1} = \gamma_i f_i$ for $1 \leq i < k$. Then $\alpha_i, \beta_i, \gamma_i \in \mathbb{Z}_{(2)}$ and

$$D(n+1) = \gamma_k\, Q(N_k, r_k)\big(\alpha_k R(N_k, r_k) + \beta_k\big). \tag{5.1}$$

Proof. By Lemma 5.3 and Proposition 4.1(ii),(iii), and since $2^{L(N_i)} = 2^{\ell_i - 1}$, we have $Q(N_i, r_i) = f_i Q(N_{i+1}, r_{i+1})$ and $R(N_i, r_i) = a_i R(N_{i+1}, r_{i+1}) + c_i$ for $1 \leq i < k$. By induction on i it follows that $Q(n+1, 0) = \gamma_i Q(N_i, r_i)$ and $R(n+1, 0) = \alpha_i R(N_i, r_i) + \beta_i$ for $1 \leq i \leq k$. Now apply Proposition 4.1(v) with $N = n+1$. □

We now reduce (5.1) modulo $P = 2^K$. Fix $K \geq 1$, let $A = \mathbb{Z}/P\mathbb{Z}$, and put

$$C = \max(K+1, 3).$$

For $0 \leq \lambda \leq C$ define $T(\lambda), T'(\lambda) \in A$ by $T(\lambda) = 2^\lambda$ and $T'(\lambda) = 2^{\lambda-1}$ for $\lambda < C$ (with $T'(0) = 0$), and $T(C) = T'(C) = 0$. For a word v put

$$\lambda(v) = \min(|v|, C), \qquad \nu(v) = [v]_2 \bmod P,$$

and for $o \in \{0,1\}$, $0 \leq \lambda \leq C$ and $\nu \in A$ put $\mu(0, \lambda, \nu) = \nu$ and $\mu(1, \lambda, \nu) = T(\lambda) - 1 - \nu$.

Lemma 5.5. *For every word v and every letter b:*

(a) $2^{|v|} \equiv T(\lambda(v))$, *and if $|v| \geq 1$ also* $2^{|v|-1} \equiv T'(\lambda(v)) \pmod P$;
(b) $[v^{(o)}]_2 \equiv \mu(o, \lambda(v), \nu(v)) \pmod P$ *for $o \in \{0,1\}$;*
(c) $\lambda(bv) = \min(\lambda(v) + 1, C)$ *and* $\nu(bv) = b\, T(\lambda(v)) + \nu(v)$.

Proof. If $|v| < C$, then $\lambda(v) = |v|$ and (a) is the definition. If $|v| \ge C$, then $|v|-1 \ge C-1 \ge K$, so both powers of two vanish modulo P. Part (b) follows from (a) and (2.1), and (c) from (a) and $[bv]_2 = b\,2^{|v|} + [v]_2$. □

Let $\Xi = \{0,1\} \times A^4$, whose elements $\xi = (o, \rho, \alpha, \beta, \gamma)$ record an orientation bit, the residue ρ of the parameter q, and the three accumulators of Proposition 5.4. We write $o(\xi) = o$ and put $\xi_{\mathrm{init}} = (0,1,1,0,1)$.

Definition 5.6. Let $\xi = (o, \rho, \alpha, \beta, \gamma) \in \Xi$.

(a) For $\chi \in \{0,1\}$, $3 \le \lambda \le C$ and $\nu \in A$ put $\mu = \mu(o, \lambda, \nu)$, $\eta = (-1)^{\mu+1}$ (well defined since P is even), $u = 1-2\rho$ (a unit of A) and $c = \eta\rho u^{-1}$, and define $\Theta(\xi; \chi, \lambda, \nu) \in \Xi$ by

$$\Theta(\xi;\chi,\lambda,\nu) = \begin{cases} \big(1-o,\ 1,\ -\alpha,\ \beta+\alpha c,\ \gamma\,\eta\,u^{2\mu+1-T(\lambda)}\big) & \text{if } \chi = 0, \\ \big(o,\ 2\rho,\ \alpha,\ \beta+\alpha c,\ \gamma\,u^{T'(\lambda)}\big) & \text{if } \chi = 1, \end{cases}$$

where powers of u with exponents in A are understood as in Lemma 2.1.

(b) For a word v with $1 \le |v| \le 2$ let $M = [v^{(o)}]_2$. If $M = 0$ put $\Omega(\xi, v) = 0$. Otherwise put

$$\Omega(\xi, v) = \gamma\, Q_{M+1}(\rho)\big(\alpha R_{M+1}(\rho) + \beta\big) \in A,$$

where $Q_N(\rho)$ and $R_N(\rho)$, for $N \in \{2,3,4\}$, are the expressions in (4.1) with q replaced by ρ, evaluated in A; their denominators $1-2\rho$ and $(1-2\rho)(1-4\rho)$ are units.

Proposition 5.7. *Let $n \ge 1$, use the notation of Definition 5.2 and Proposition 5.4, and let $\xi_i \in \Xi$ be the reduction of $(o_i, q_i, \alpha_i, \beta_i, \gamma_i)$ modulo P. Then $\xi_1 = \xi_{\mathrm{init}}$,*

$$\xi_{i+1} = \Theta\big(\xi_i;\ b_{i+1} \oplus o_i,\ \lambda(b_i \cdots b_t),\ \nu(b_i \cdots b_t)\big) \qquad (1 \le i < k),$$

and

$$D(n+1) \equiv \Omega(\xi_k,\, b_k \cdots b_t) \pmod{P}.$$

Proof. Let $1 \le i < k$, and write $\lambda_i = \lambda(b_i \cdots b_t)$, $\nu_i = \nu(b_i \cdots b_t)$ and $\mu_i = \mu(o_i, \lambda_i, \nu_i)$. Since $\ell_i \ge 3$ and $C \ge 3$ we have $\lambda_i \ge 3$. By Lemmas 5.3 and 5.5(b), $N_i - 1 \equiv \mu_i \pmod{P}$, so $\eta_{N_i} = (-1)^{\mu_i+1}$. Since $L(N_i) = \ell_i - 1$,

$$e(N_i) = 2(N_i-1) + 1 - 2^{\ell_i} \equiv 2\mu_i + 1 - T(\lambda_i), \qquad 2^{\ell_i - 1} \equiv T'(\lambda_i) \pmod{P}$$

by Lemma 5.5(a). As u_i is odd, Lemma 2.1 shows that the residues of $u_i^{e(N_i)}$ and $u_i^{2^{\ell_i-1}}$ depend only on these exponents modulo P. Since reduction $\mathbb{Z}_{(2)} \to A$ is a ring homomorphism, comparing the updates of o_i and r_i in Definition 5.2 (so that q_{i+1} is 1 or $2q_i$) and the updates of $\alpha_i, \beta_i, \gamma_i$ in Proposition 5.4 with Definition 5.6(a) gives the formula for ξ_{i+1}.

At the final stage, $N_k = M + 1$ with $M = [(b_k \cdots b_t)^{(o_k)}]_2 \in \{1,2,3\}$ by Lemma 5.3; note that M is determined by the at most two digits $b_k \cdots b_t$ and by o_k, not merely by a residue. By Proposition 4.1(i), $Q(N_k, r_k)$ and $R(N_k, r_k)$ are rational functions of q_k with odd denominators, whose reductions are $Q_{M+1}(\rho_k)$ and $R_{M+1}(\rho_k)$. Reducing (5.1) modulo P completes the proof. □

The input to Proposition 5.7 at stage i consists of the state ξ_i, one further digit, and the pair (λ, ν) of the suffix $b_i \cdots b_t$, all of which range over finite sets. The suffix data are not available to an automaton that reads $\mathrm{bin}(n)$ from the left; this is why the automaton below reads from the right.

6. The automaton

We keep K, P, A, C, Ξ, Θ and Ω from Section 5.

Definition 6.1. For a nonempty word v define a map $\Phi_v\colon \Xi \to A$ by recursion on $|v|$. If $|v| \le 2$, put $\Phi_v(\xi) = \Omega(\xi, v)$. If $|v| \ge 3$, write $v = bv'$ with a letter b, let v'_1 be the first letter of v', and put

$$\Phi_v(\xi) = \Phi_{v'}\big(\Theta(\xi;\ v'_1 \oplus o(\xi),\ \lambda(v),\ \nu(v))\big).$$

We also define Φ_ε to be the zero map.

Lemma 6.2. *For every $n \ge 1$, $\Phi_{\mathrm{bin}(n)}(\xi_{\mathrm{init}}) \equiv D(n+1) \pmod P$.*

Proof. Let $\mathrm{bin}(n) = b_1 \cdots b_t$ and let $\xi_1, \dots, \xi_k$ be as in Proposition 5.7. For $i < k$ the word $b_i \cdots b_t$ has length at least 3, so Definition 6.1 and Proposition 5.7 give $\Phi_{b_i \cdots b_t}(\xi_i) = \Phi_{b_{i+1} \cdots b_t}(\xi_{i+1})$. Hence $\Phi_{\mathrm{bin}(n)}(\xi_{\mathrm{init}}) = \Phi_{b_k \cdots b_t}(\xi_k) = \Omega(\xi_k, b_k \cdots b_t)$, which is congruent to $D(n+1)$ by Proposition 5.7. □

For a word v containing the letter 1, let v° be the word obtained by deleting the leading zeros of v, so that $v^\circ = \mathrm{bin}([v]_2)$. Let $\pi(v)$ be the prefix of v of length $\min(|v|, 2)$, and put

$$\mathrm{out}(v) = \begin{cases} \Phi_{v^\circ}(\xi_{\mathrm{init}}) & \text{if } v \text{ contains the letter 1,} \\ 0 & \text{otherwise.} \end{cases}$$

Finally, define

$$\mathfrak{s}(v) = \big(\lambda(v),\ \nu(v),\ \pi(v),\ \Phi_v,\ \mathrm{out}(v)\big).$$

Then $\mathfrak{s}(v)$ lies in the finite set

$$\mathcal{S} = \{0, \dots, C\} \times A \times \{\text{words of length at most } 2\} \times A^{\Xi} \times A.$$

Lemma 6.3. *There is a map $F\colon \{0,1\} \times \mathcal{S} \to \mathcal{S}$ such that $\mathfrak{s}(bv) = F(b, \mathfrak{s}(v))$ for every word v and every letter b.*

Proof. We show that each component of $\mathfrak{s}(bv)$ is determined by b and $\mathfrak{s}(v)$. For $\lambda(bv)$ and $\nu(bv)$ this is Lemma 5.5(c), and $\pi(bv)$ is the prefix of length $\min(|bv|, 2)$ of $b\,\pi(v)$. Note that $|bv| \le 2$ if and only if $\lambda(v) \le 1$. If $\lambda(v) \le 1$, then $bv = \pi(bv)$ and $\Phi_{bv} = \Omega(\cdot, \pi(bv))$. If $\lambda(v) \ge 2$, then $|bv| \ge 3$, the first letter of v is the first letter of $\pi(v)$, and by Definition 6.1

$$\Phi_{bv}(\xi) = \Phi_v\big(\Theta(\xi;\ \pi(v)_1 \oplus o(\xi),\ \lambda(bv),\ \nu(bv))\big).$$

Finally, $\mathrm{out}(bv) = \Phi_{bv}(\xi_{\mathrm{init}})$ if $b = 1$, since then $(bv)^\circ = bv$; and $\mathrm{out}(bv) = \mathrm{out}(v)$ if $b = 0$, since then bv contains a 1 exactly when v does, and in that case $(bv)^\circ = v^\circ$. These formulas define $F(b, \sigma)$ for σ in the set $S = \{\mathfrak{s}(v) : v \text{ a word}\}$ of states that occur; on the remaining elements of $\mathcal{S}$ we extend F arbitrarily, for instance by $F(b, \sigma) = \sigma$. □

Proof of Theorem 1.1. Fix $K \ge 1$ and let $S = \{\mathfrak{s}(v) : v \text{ a word}\} \subseteq \mathcal{S}$, a finite set. By Lemma 6.3, the map $\delta(\mathfrak{s}(v), b) = \mathfrak{s}(bv)$ is well defined on $S \times \{0,1\}$. Consider the 2-DFAO $M = (S, \delta, \mathfrak{s}(\varepsilon), A, \tau)$, where τ is the projection onto the last coordinate. By induction on $|w|$ we have $\hat\delta(\mathfrak{s}(v), w) = \mathfrak{s}(wv)$ for all words v, w: indeed $\hat\delta(\mathfrak{s}(v), w'b) = \hat\delta(\mathfrak{s}(bv), w') = \mathfrak{s}(w'bv)$. In particular $\hat\delta(\mathfrak{s}(\varepsilon), w) = \mathfrak{s}(w)$.

Let $n \ge 0$ and let w be a word with $[w]_2 = n$. If $n \ge 1$, then w contains a 1 and $w^\circ = \mathrm{bin}(n)$, so $\tau(\mathfrak{s}(w)) = \Phi_{\mathrm{bin}(n)}(\xi_{\mathrm{init}}) \equiv D(n+1) \pmod P$ by Lemma 6.2.

If $n = 0$, then w contains no 1 and $\tau(\mathfrak{s}(w)) = 0 = D(1)$. Hence M generates $(D(n+1) \bmod 2^K)_{n\geq 0}$. □

Remark 6.4. Let $a(n) = D(n+1) \bmod 2^K$. For $i \geq 0$ and $0 \leq j < 2^i$ let v_j be the word of length i with $[v_j]_2 = j$. Then $[\mathrm{bin}(n)v_j]_2 = 2^i n + j$, hence

$$a(2^i n + j) = \tau\big(\hat{\delta}(\mathfrak{s}(v_j), \mathrm{bin}(n))\big) \qquad (n \geq 0),$$

so the sequence $(a(2^i n+j))_{n\geq 0}$ depends on (i, j) only through the state $\mathfrak{s}(v_j)$. Thus the 2-kernel of $(a(n))_{n\geq 0}$ has at most $|S|$ elements. This gives Theorem 1.1 directly in the formulation by finite 2-kernels; the equivalence with automata reading the most significant digit first is [3, Section 5.2].

Remark 6.5. The construction bounds $|S|$ by $(C+1) \cdot P \cdot 7 \cdot P^{2P^4} \cdot P$; this upper bound is very large already for $K = 2$. We do not determine minimal automata. For comparison, the automata found experimentally in [11] for $K = 2$ and $K = 3$ have 12 and 83 states.

7. Further remarks

7.1. The determinants are not 2-regular.

Proposition 7.1. *For $m \geq 2$ we have $|D(2^m)| \geq 3^{2^{m-2}}$. Consequently the sequence $(D(N))_{N\geq 1}$ is not 2-regular.*

Proof. By [11, Proposition 4.5], for $m \geq 2$,

$$D(2^m) = \Big(\sum_{j=1}^{m} \frac{2^{j-1}}{2^j - 1}\Big) \prod_{j=1}^{m} (2^j - 1)^{2^{m-j}}.$$

All terms are positive, the sum is at least its first term 1, and the product is at least its factor $3^{2^{m-2}}$ with $j = 2$. A 2-regular sequence of complex numbers is $O(N^c)$ for some constant c [2, Theorem 2.10], which is incompatible with $|D(N)| \geq 3^{N/4}$ for $N = 2^m$. □

Accordingly, the proof of Theorem 1.1 propagates only residues, through the normalized pair (Q, R) of Proposition 4.1, and never the integers $D(N)$ themselves.

7.2. Other moduli.

The reduction in Section 5 rests on the fact that the denominators $1 - 2q$ and $1 - 4q$ in Proposition 4.1, for q a power of two, are units modulo 2^K. For a modulus with an odd prime factor p this fails in general, since $1 - 2q \equiv 0 \pmod p$ when $2q \equiv 1 \pmod p$. The construction also reduces the exponents in (4.2) and (4.3) modulo 2^K itself (Lemma 2.1); modulo another integer m, exponents cannot in general be reduced modulo m. For instance, for $m = 3$ and $q = 1$ we have $u = -1 \equiv 2 \pmod 3$, and the exponents 0 and 3 agree modulo 3, whereas $2^0 = 1$ and $2^3 \equiv 2 \pmod 3$. We do not address such moduli. By [11, Corollary 4.16], for every odd m the set of n with $m \mid D(n)$ has density 1.

7.3. Finite computations.

The supplementary file `verification.zip` accompanying this paper contains a Python script `verify_r57.py`, which uses only the standard library, its output `verification_results.json`, and a description `README-verification.txt`. These finite computations are not part of the proof of Theorem 1.1. The script computes $g(N, 2^{-r})$ and $h(N, 2^{-r})$ from their defining matrices by integer Bareiss elimination and finds that $Q(N, r)$ and $R(N, r)$

agree with the values given by (4.1)–(4.3), and satisfy Proposition 4.1(iv), for $2 \le N \le 40$ and $0 \le r \le 8$. It finds that the iteration of (4.2)–(4.3) modulo 2^K agrees with $D(N) \bmod 2^K$ for $2 \le N \le 40$ and $1 \le K \le 12$, and that the formula of Proposition 5.7, with the suffix data computed by Lemma 5.5(c), agrees with this iteration for $0 \le n \le 5000$ and $1 \le K \le 10$, as well as for selected inputs with up to 2048 binary digits. The values $D(1), \dots, D(16)$ obtained agree with those listed in the introduction. The detailed record of the run is contained in `verification_results.json`.

Use of AI-assisted tools

AI-assisted tools were used in the research and preparation of this manuscript.

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